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MILITARY SPECIFICATION

INTERROGATOR SET, AN/UPX-24(V)

1. SCOPE

1.1 Scope. This specification covers the requirements for performance, technical design, and quality assurance for a Central Identification, Friend or Foe (IFF) (Air Traffic Control Radar Beacon System (ATCRBS)) shipboard system.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards, and handbooks. The following Specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

Military

MIL-S-901	Shock Tests, H.I. (High-Impact), Shipboard Machinery, Equipment and Systems Requirments For
MIL-E-15090	Enamel, Equipment, Light Gray (Formula No. 111)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Systems Command (AIR-5516), Washington, DC 20361-5510, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5895
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MIL-C-915	Cable and Cord Electrical, For Shipboard Use General Specification For
MIL-C-5015	Connectors, Electrical AN Type
MIL-E-5400	Electronic Equipment, Aerospace General Specification For
MIL-C-5541	Chemical Conversion Coatings on Aluminum And Aluminum Alloys
MIL-Q-9858	Quality Program Requirements
MIL-E-13777	Cable, Special Purpose, Electrical, General Specification For
MIL-E-16400	Electronic, Interior Communication and Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-A-28737	Antenna, IFF, AS-1288(), (3 meters) (10 feet) AS-2189 (1.8 meters) (6 feet)
MIL-I-28763	Interrogator Set, AN/UPX-27
MIL-P-28809	Circuit Card Assemblies, Rigid, Flexible and Rigid - Flex
MIL-P-55110	Printed Wiring Boards, General Specification for

STANDARDS

MILITARY

MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations of Shipboard Equipment (Type I-Environmental and Type II-Internally excited)
MIL-STD-242	Electronic Equipment Parts Selected Standards Switches
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements For the Control of Electromagnetic Interference
MIL-STD-469	Radar Engineering Design Requirements Electromagnetic Compatibility
MIL-STD-470	Maintainability Program For Systems And Equipments
MIL-STD-471	Maintainability Verification Demonstration/Evaluation
MIL-STD-781	Reliability Testing for Engineering Development, Qualification, and Production
MIL-STD-965	Parts Control Program
MIL-STD-1364	General Purpose Electronic Test Equipment
MIL-STD-1397	Input/Output Interfaces, Standard Digital Data, Navy Systems
MIL-STD-1520	Corrective Action And Disposition System For Non-Conforming Material
MIL-STD-1535	Supplier Quality Assurance Program Requirements
MIL-STD-1686	Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices) (Metric)
MIL-STD-2000	Standard Requirements for Soldered Electrical and Electronic Assemblies

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FEDERAL

Fed-STD-595	Colors Used in Government Procurement

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

DEPARTMENT OF DEFENSE

DoD AIMS 64-402	RT/Receiver Transmitter Test Item.
DoD AIMS 64-600	Antenna Electronically Scanned
DoD AIMS 65-619	Decoder Group AN/UPA-() Variable Configuration
DoD AIMS 65-620	Interrogator Set AN/UPX-23
DoD AIMS 65-1000	Technical Standard For AIMS

(Copies of DOD AIMS Specifications are available from WR-ALC/LYLR-AIMS Robins Air Force Base, GA 31098)

2.1.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The central IFF system shall be in accordance with MIL-E-16400 to the extent specified herein.

3.1.1 Performance Requirements. The central IFF system shall be capable of operating independently of interconnection with other radar or electronic systems except to utilize existing display methods as appropriate. The purpose of this development is to provide an electronic system with the capability to expand or reduce the number of functional units to permit an adaptability to match the different functional aspects of the ship in which the system is installed. Therefore, there will be certain basic elements common to all system configurations while additional specialized elements will be used specifically for tactically different and more complex system configurations.

3.1.2 Typical Shipboard Systems. The central IFF system shall be required to meet various shipboard system configurations. Some ships will have only a few (4 to 8) operating positions and the radar complement may be only a surface search and an air search radar system. Other shipboard configurations may have many (14 to 22) operating positions and have radar systems for surface search, two air search and an air traffic control system. Some of the more complex ship systems will also have automatic or manual Naval Tactical Data System (NTDS) configurations. The central IFF system shall be designed for all shipboard configurations and a certain equipment flexibility shall be included in the design so that the simpler systems are not required to have an IFF system complexity beyond their needs.

3.1.3 Operator Control Functions. The operator at each of the multiple positions shall have an independent choice of the IFF system operation to suit the operator's needs. The operator shall be able to control the data displayed on the indicator (except in certain uses of Mode override) without considerations of the controls being placed upon the IFF system by the other operators. The operator shall be able to challenge in any combination of Modes 1,2, 3/A and C and use Mode 4 on an override basis. The operator shall have independent choice on the altitude layer filtering at his position; shall be able to select the area gate for active readout of the selective identification feature (SIF) codes and Mode C; and to choose the passive decodes for Modes 1, 2, and 3/A. The operator shall be able to control the displayed IFF information to provide all targets (bracket decoding); to display only those targets conforming to the criteria of the passive decodes;

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to display all targets and have distinctive marking on all targets conforming to the criteria of the passive decodes; and to display emergency signals. Passive code display of special replies such as I/P and X pulse position shall also be under the control of the operator. Additionally, the operator shall be able to control the automatic challenges to be produced only during the sector or sectors of interest. At a master operator position, (one only) control of the RF power output shall be available so that only the power necessary for the range of operation shall be omitted. Various power levels may exist for a 360° rotation.

3.1.4 Typical Displays. There shall be a special numeric readout indicator associated with the control panel of the operator. The display shall provide a displayed arc for bracket decoding, a distinctive slash mark to the arc when a Mode 4 reply conforms to certain acceptance criteria, and a distinctive display stretching for each passive decode. Other display symbols shall include non mode C, I/P on X pulse, and emergency.

3.1.5 Functional Block Diagram. The variety of system configurations that shall be considered are shown in FIGURES 1, 2 and 3.

3.2 Basic System Elements. The basic system elements shall consist of existing AIMS IFF equipments insofar as is practical.

3.2.1 Basic System Elements. The basic system elements shall be as specified in (a) through (c):

a. Processor - Controller (PC), CP-1273/UPX-24

b. Control Monitor (CM), C-10065/UPX-24

c. Control Indicator (CI), C-10064/UPX-24

3.2.2 Processor Controller (PC). The PC shall degarble, defruit, decode, target validate, and indicate target positions. Detailed characteristics of the PC are specified in 3.8 through 3.8.13.

3.2.3 Control Indicator (CI). The control box shall provide operator control of the system at a multiple of operating positions, each independently of the other. The indicator shall provide a numeric display showing the mode, code, and altitude reporting of the reply under investigation.

3.2.4 Control Monitor (CM). The CM shall serve as the master control position for the AN/UPX-24 system. The CM shall provide timing and control functions necessary to allow real-time sampling of the controls and switches by the PC. The CM shall provide the switches, controls, and indicators to perform the following functions:

a. Provide sector control for three sectors for which each sector can be independently controlled for start and stop azimuth, specify modes of interrogations, radio frequency (rf) output power levels, and the pulse repetition time (PRT).

b. Monitor and control AIMS Interrogator Set functions for Interrogation side lobe suppression (ISLS) fault indication, selection of gain time control (GTC) lows or short gate, and high voltage overload indication, and reset control.

c. Monitor and control the KIR-1A Mode 4 computer functions.

d. Provide Navy Tactical Data System (NTDS) buffer range control for a minimum and a maximum range selection.

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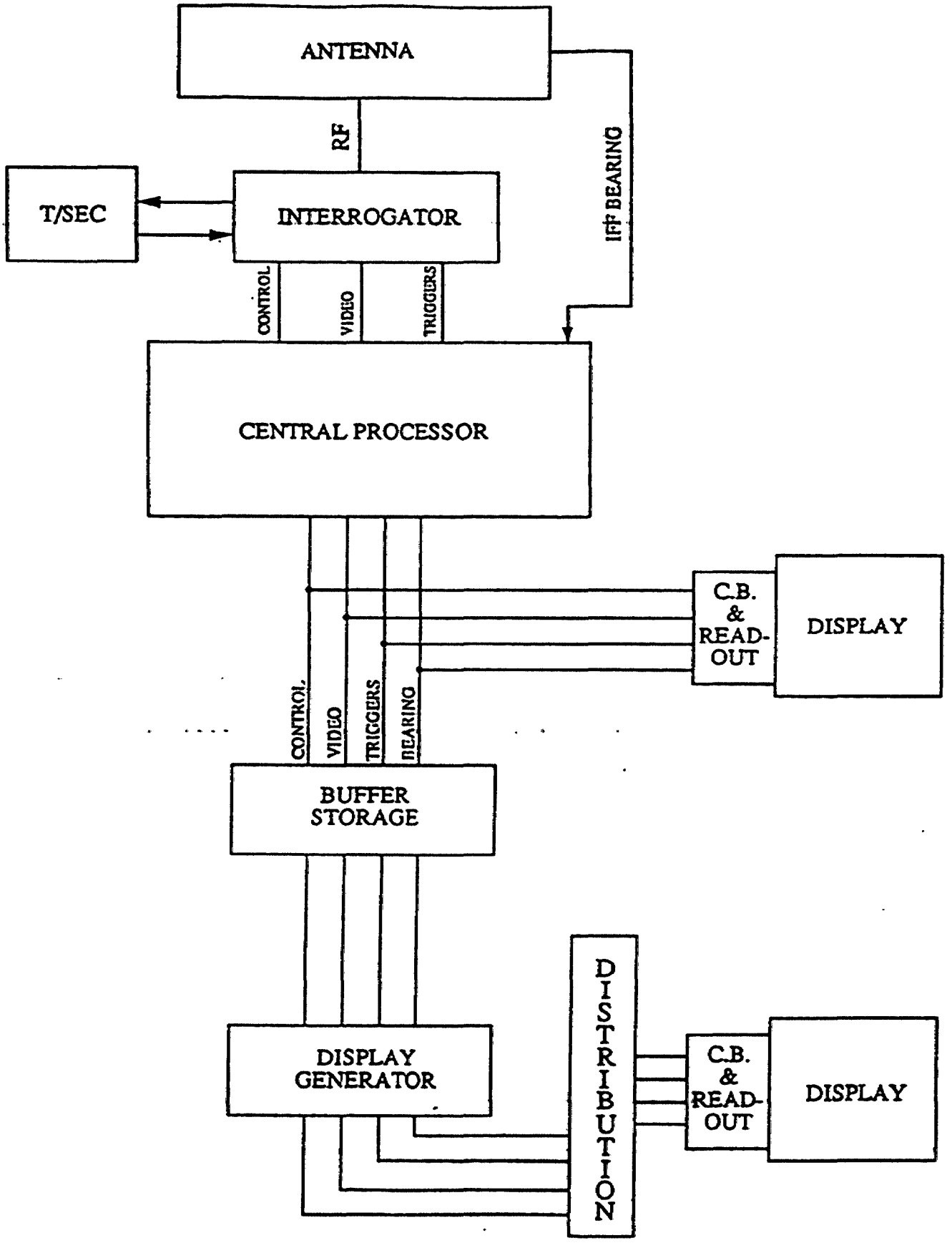


FIGURE 1. Manual or automatic system (non-NTDS)

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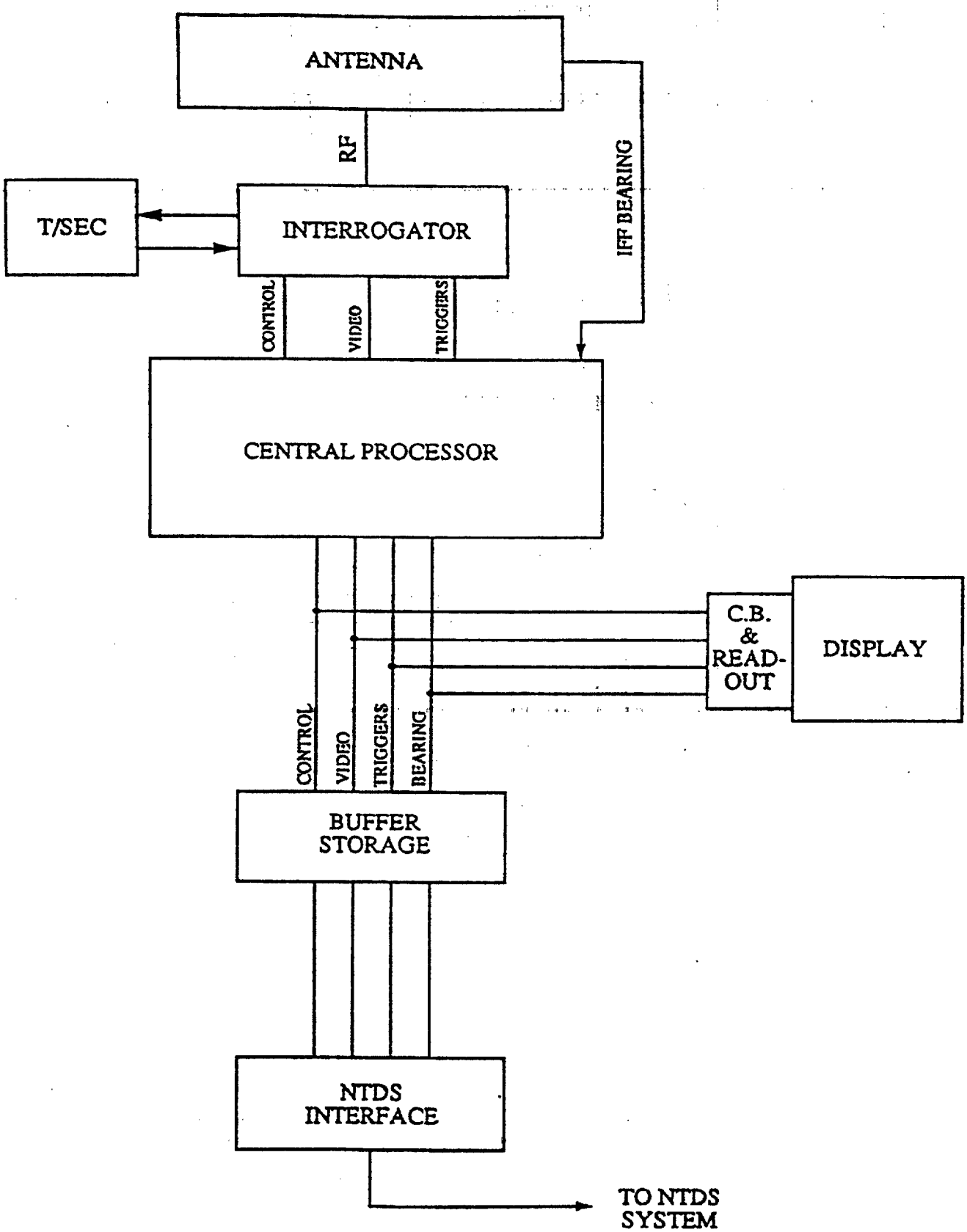


FIGURE 2. NTDS System

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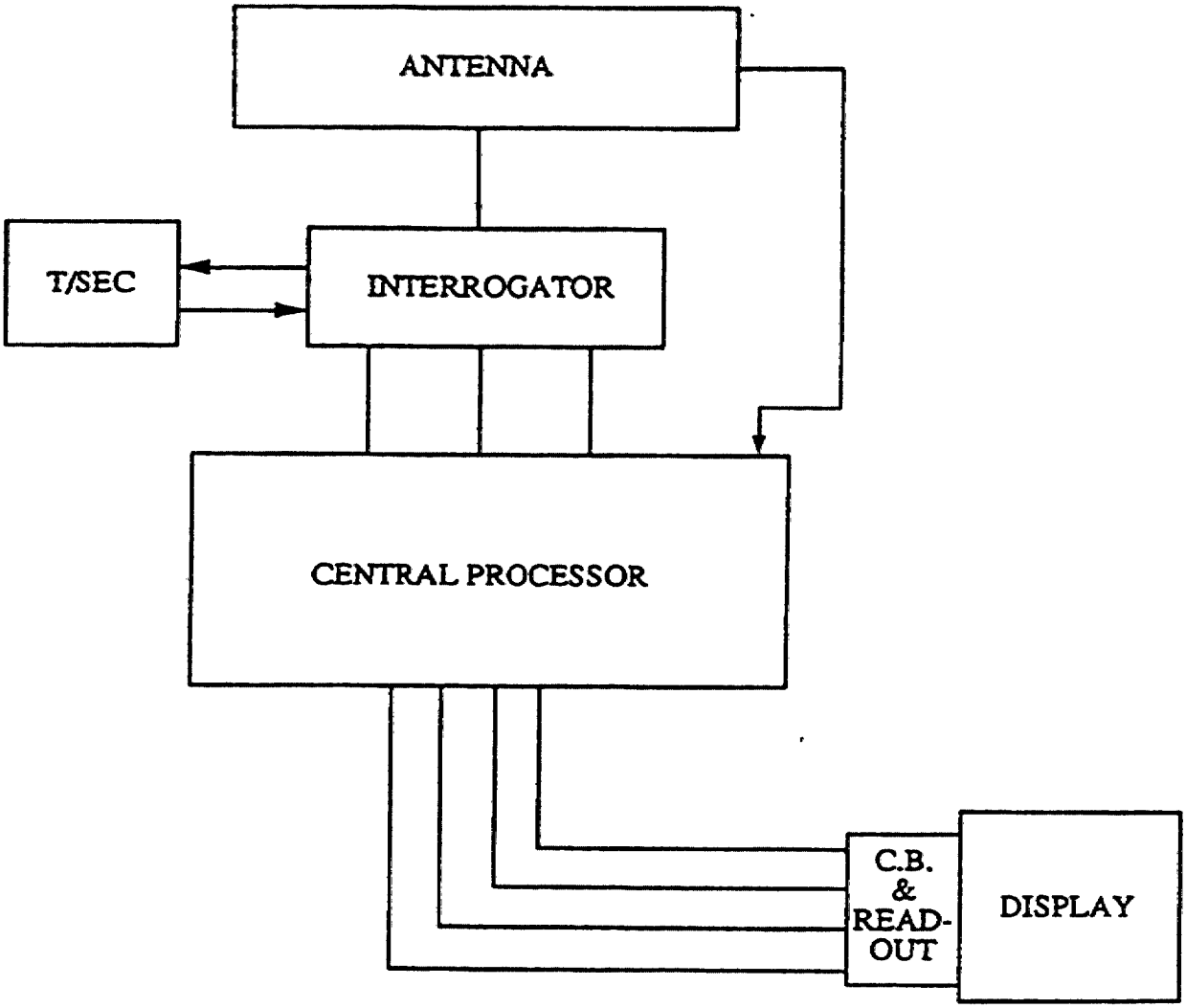


FIGURE 3. Simple Manual System

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e. Monitors alarm functions of the PC for processor overload, radar range zero pulse failure, radar pretrigger pulse failure, and processor overtemperature.

f. Monitors the following PC functions, system failures, and operations in bypass mode of operation.

3.3 Specialized System Functions. The specialized system functions shall consist of a buffer storage , a display generator, an NTDS interface , and a data distribution system. Additional specialized elements such as other types of control boxes and readout indicators may be required to operate with the different design features of the specialized elements; however, these additional specialized elements shall be kept to a minimum.

3.3.1 Buffer Storage. Buffer storage shall be provided for the more complex system configurations. When the central IFF system is configured to supply information only to the NTDS system (manually operated or automatically processed), the buffer storage shall hold and supply data in accordance with programmed or manual commands of the NTDS system supplied via the NTDS interface. When the central IFF system is configured to be a combined automatic and manual system (for ships without NTDS), the buffer storage unit shall supply decoded data via the display generator for the operation and the display of the IFF information. This buffer storage shall include target position information as well as the specific code information of each target relay.

3.3.2 NTDS Interface. The interface equipment shall accept NTDS program requests, control the operation of the IFF system in accordance with the NTDS program request, and translate the IFF information of target position and coded replies into suitable language for use by the NTDS system.

3.3.3 Display Generator. The display generator shall convert the stored data to appropriate display signals. The minimum performance requirements are specified in (a) through (f):

a. Bracket decode of all targets

b. Passive decode on selected targets

c. I/P, X position, and emergency decodes

d. Numerical readout of the SIF codes

e. Numerical readout of Mode C (altitude) codes

f. A Mode 4 symbol indicating that an evaluation criteria was met.

3.3.4 Distribution System. The distribution of the control and display information shall be a direct wiring technique.

3.3.5 Special Type Control and Display Filter. The special type control and display filters are special system elements in that control, and display via the buffer storage and display generator can be quite different from similar functions from the basic element counterparts. The control functions shall remain essentially the same, but the readout and display data shall be obtained more from filtering out the unwanted data rather than from generating readout and display from chosen data.

3.4 Design Requirements.

3.4.1 General Design. The equipments shall operate compatibly with the AIMS IFF systems of all services. DoD AIMS 65-1000 shall be the governing document for determining the technical characteristics and performance

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requirements relative to this compatibility. In addition, the equipment shall conform to the full requirements for operation in a shipboard environment including nuclear air blast, stack gas temperatures, salt laden air, and continued pitch and roll conditions.

3.4.2 Operational System Requirements. The central IFF system shall conform to the performance requirements specified in (a) through (e).

a. A target detection probability of 98 percent (or better) on each of the modes replying during a single antenna illumination

b. A code validation probability of 95 percent (or greater) on each of the modes replying during a single antenna illumination period

c. A false target detection of 2 percent (or less) on each of the modes replying during a single antenna illumination period

d. A split target probability of 5 percent (or less) on each of the modes replying during a single antenna illumination period

e. Full performance in a fruit rate environment of up to 5000 asynchronous target replies per second

3.4.3 Temperature Requirements. Equipments located in rooms, compartments, and similar sheltered environments shall conform to Class 4 of MIL-E-16400.

3.4.4 Humidity. Equipments exposed to the weather shall be fog-proof and equipments in sheltered areas such as rooms shall be dripproof in accordance with MIL-STD-108.

3.4.5 Parts Control Program. Parts to be incorporated in the equipment shall be selected and controlled in accordance with MIL-STD-965, procedure I.

3.4.6 Nonstandard Parts Approval. Nonstandard parts shall not be used in the system unless approved by the procuring activity.

3.4.7 Solid State Parts. Maximum use shall be made of solid state parts in the central IFF system. Thermionic and ionizing tubes shall be used only where no solid state devices adequate for the purpose are available, or where such use of solid state devices would result in excessive cost or greater circuit complexity. Solid state components capable of working over a broad range of temperature shall be used wherever their use is technically feasible, and cost effective.

3.4.8 Cables and Connectors. All power and control cabling shall be as specified in MIL-C-13777 (portable use) and MIL-C-915. Coaxial cables and connectors shall be selected from MIL-STD-242. Connectors for multiple conductor cables interconnecting major units shall be in accordance with MIL-C-5015. Mating connectors shall be furnished with all connectors, receptacles, or plugs.

3.4.9 Electrical Design. The electrical design of all equipment shall be in accordance with MIL-E-16400, except as otherwise specified herein. The equipment shall be designed and fabricated for a minimum loss of operational capability as functions are lost or become inoperative due to electrical, electronic, or physical failures. The equipment shall also be designed with as small a number of different physical units and subassemblies as is feasible.

3.4.10 Overload Protection. Protective devices shall be provided within the equipment for protection from damage due to overload and excessive heating in accordance with MIL-E-16400, Table I and MIL-STD-454, Requirement 39. If fuses are used, blown fuse indicators shall be provided, and spare

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fuses shall be mounted in the equipment, preferably adjacent to the corresponding active fuse.

3.4.11 Identification of Conductors. Conductors shall be identified in accordance with MIL-STD-454, Requirement 20.

3.4.12 Fail-Safe-Provisions. The design of the equipment shall be fail-safe. In the event of a failure, no additional or avalanche failures shall occur. The design of the equipment shall preclude damage to the modules due to incorrect or accidental turn-on or turn-off.

3.4.13 Maintenance Design. The equipment shall be designed to provide ease of maintenance in accordance with MIL-E-16400, Section 3. Maintenance procedures shall follow the concepts outlined in MIL-STD-470, Task 206. Extension cards or adapter cables necessary for servicing shall be supplied with the equipment.

3.4.14 Test, Measurement, and Diagnostic Equipment (TMDE). Test provisions for the equipment shall conform to MIL-STD-454, Requirement 32, except as specified in 3.4.14.1 through 3.4.14.5.

3.4.14.1 Test Point Criteria. Except for shop testing, test points shall be accessible with the removal of no more than one cover panel, while the equipment is operational, and without removal of cables or connectors. The use of extender boards is permitted to allow access to test points on printed circuit (PC) cards. Test points shall be provided for the requirements specified in (a) through (d) (multiple use of test points is permitted).

a. Signal quality and circuit performance. Test points, test jacks, or both, shall be provided to permit monitoring of signals.

b. Alignment. Test points, test jacks, or both, shall be provided to monitor the intermediate circuit affected by the alignment. Where TMDE is used to align or adjust a system unit, assembly, or subassembly, circuitry shall be designed so that a technician can see and operate the TMDE while making adjustments and probing the circuit under test.

c. Fault isolation. Test points, test jacks, or both, shall be provided to permit the measurement of signals to determine satisfactory operation or malfunction of the unit.

d. Shop testing. Shop testing of assemblies and subassemblies shall be performed with the assemblies and subassemblies removed from the prime equipment. Test points may be provided so that the assembly can be:

    1. Activated (such as direct current (DC)) and dynamic signals applied.

    2. Checked for quality of operation.

    3. Fault isolated to the shop replaceable items.

3.4.14.2 General Purpose Electronic Test Equipment (GPETE). GPETE for use at all levels of maintenance shall be selected from the standard and substitute standard GPETE listed in MIL-STD-1364.

3.4.14.3 Special Tools. Special tools shall be supplied in accordance with MIL-STD-454, Requirement 63. Special tools shall conform to the storage and environmental operating conditions specified for the prime equipment.

3.4.14.4 Built-In Test Equipment (BITE). BITE requirements shall be as follows:

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a. Levels. BITE shall be provided for the following levels:

1. Level A. Level A BITE shall provide a GO-NO-GO indication of performance and quality of the system.

2. Level B. Level B BITE shall provide a GO-NO-GO determination of quality and performance of the assembly.

3. Level C. Level C BITE shall indicate and isolate the failure to the subassembly or subassemblies.

b. Activation.

1. Levels A and B BITE activation shall be automatic and shall not require operator initiation.

2. Level C BITE activation may be automatic or manual.

3. All BITE functions shall be generated during the signal processing cycle.

c. GO-NO-GO indicators. All BITE GO-NO-GO indicators shall be covered or uncovered. Indicators shall be made visible by the removal of not more than one cover plate or panel.

1. Indicators for level A BITE for ship, shore, and vehicular installations shall be lights, meters, or other indicators installed on the system control panel.

2. Indicators for level B BITE shall be provided on the front panel of each assembly.

3. Indicators for level C BITE shall be provided on the top surface of each subassembly.

d. Dependability

1. For system failures which affect the functional operation of the system, the probability of automatic fault detection using level A BITE shall be 0.84.

2. For detectable failures which lie within subassemblies, the probability of fault isolation (PFI) shall be 0.95 using level B BITE.

3. For detectable failures which lie within functional PC cards, the PFI to n subassemblies shall be:

PFI	n
0.9	3
0.8	2
0.7	1

3.4.14.5 Computer Designated Test Target. A test target shall be provided via the computer interface at a range and bearing designated by the computer input request.

3.4.15 Parts Identification by Reference Designation. In order to facilitate maintenance, each part assembled in the IFF interrogator set shall be identified by an appropriate reference designation in accordance with MIL-E-16400, Section 3 and MIL-STD-454, Requirement 67.

a. CI and CM. Reference designations on the CI and CM are permitted to be hidden by harness routing.

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b. Circuit boards. Circuit board component parts containing values and part numbers shall be visible whenever practical.

c. Wireboards. On those wire boards where space does not permit the reference designations to be printed on the boards, the requirement to print a reference designation is deleted for that board only. Decision as to printing/non-printing shall be on a board-by-board basis, with mutual agreement between the Government and Litton Systems, Data Systems Division.

d. Other designations. Reference designations S21, S22, S50 through S57, may be omitted on the back of the CM panel Part Number 169128. Also, reference designations for S5 through S14, S17 through S25, and S29 through S38 may be omitted on the back of the CI panel Part Number 169119.

e. Printed boards. Where space is at a premium, printed circuit boards are not required to have reference designations as called for in Section 3 of MIL-E-16400.

3.4.16 Modular Construction. As specified in the Section 3, accessability paragraph of MIL-E-16400 and MIL-STD-454, Requirement 36, the equipment shall employ unitized construction and modular design to provide easily replaceable assemblies and plug-in subassemblies for standardization of similar circuits and to facilitate maintenance by rapid substitution. The design of plug-in assemblies shall consider utilizing available assemblies and subassemblies whenever economic or logistic benefits can be achieved. Each assembly and each plug-in subassembly shall provide a complete circuit function(s) with readily accessible test points at input and output areas. Plug-in subassemblies shall be of the printed circuitry type which can be removed without need of unsoldering and which can be easily repaired.

3.4.17 Shock. The system shall conform to the Grade A, Class I, Type A lightweight requirements of MIL-S-901.

3.4.18 Vibration. The system shall conform to the vibration requirements of MIL-STD-167-1, Type I and Section 3 of MIL-E-16400.

3.4.19 Inclination. The system shall conform to the 45 degree inclination requirements of MIL-E-16400, Section 3, for surface platform installations.

3.4.20 Service Environmental Requirements. The equipment shall be capable of continuous operation during any combination of the service environmental conditions specified in (a) through (g):

a. Ambient air temperature, -62°C to +71°C in accordance with Class 4 of MIL-E-16400.

b. Relative humidity, 0 to 95 percent

c. Salt laden air as encountered in a shipboard environment

d. Sand and dust particles 10 microns to 50 microns in size

e. Winds up to 100 miles per hour

f. Air blast up to 10 per square inch peak overpressure

3.4.21 Service Operating Conditions. In combination with the service environmental requirements, the equipment shall be simultaneously capable of continuous operation under and combination of the supplementary service operating conditions specified in (a) through (e):

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a. Power supply safety factor

b. RF duty cycle of zero to 1 percent

c. Voltage standing wave ratio (VSWR) of up to 3.5 decibel (dB) (antenna mismatch), that is, 1.5 VSWR to 1 VSWR

d. Line voltage variations

e. Primary supply frequency variations

3.4.22 Storage Operation. The equipment shall be capable of satisfactory operation after being stored for a period not exceeding 1 year at storage room ambient temperature ranges from -62°C to +71°C.

3.4.23 Electromagnetic Interference Emission and Susceptibility and Equipment and Systems Self-Compatibility. The equipment shall conform to MIL-STD-461, and shall be self-compatible. The equipment is designated Class A4 in accordance with MIL-STD-461. In addition to compliance with MIL-STD-461, the equipment shall conform to the engineering design criteria of MIL-STD-469. In the event of a conflict between the requirements of MIL-STD-461 and MIL-STD-469, the latter shall govern.

3.4.24 Reliability. The AN/UPX-24(V) shall have a lower test mean-time-between-failure (theta$_1$, as defined by MIL-STD-781) of 1000 hours. Reliability prediction shall assume a shipboard environment with +65°C ambient temperature for above deck components and a +50°C ambient temperature for below deck components.

3.4.25 Maintainability. As a result of any performance failure, the system shall be capable of repair (including tuning or alignment) and ready for operation within 30 minutes. Equipment replacement is not acceptable. Module or subassembly replacement will be permitted. The mean-time-to-repair (MTTR) shall be no more than 30 minutes in accordance with MIL-STD-471.

3.4.25.1 Quantitative Maintainability Requirements. The system shall have a MTTR of not more than 30 minutes, and a maximum repair time of not more than 90 minutes at the 95th percentile and a 90 percent confidence level.

3.4.26 Terminal Designation. Terminal identification shall be provided for all internal connectors and for all external connectors used to connect units of the equipment to units of another equipment. All connections to numbered terminal board positions shall be included in appropriate diagrams contained in technical manuals and installations drawings. All external connectors connected to coaxial transmission lines shall be identified with the impedance of the transmission line.

3.4.27 Test Points. Adequate test points shall be brought out to permit isolation of a malfunction to the smallest replaceable subassembly therein if at all practical. Test points shall be identified by alphanumeric marking to facilitate location except that color coding shall be employed where space does not permit alphanumeric marking. Test points located on circuit boards and extension cards shall be marked in groups of five, by alternating bands of identifying colors. In addition, for all sockets where space and design limitations prevent the use of test points, an adapter socket, extension card, or cable shall be provided to aid in making voltage and pattern measurements that may be required in repair or maintenance of the equipment. The requirements specified herein shall provide essential test points; however, all signal input and output test points are required on all plug-in subassemblies. Test points for all DC supply voltage (including modulator high voltage) shall be provided, and shall be plainly labeled and conveniently located for ease of servicing.

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3.4.28 Coatings. The CI and CM shall be painted gray in accordance with Fed-STD-595 Color No. 36231. The PC shall be painted gray in accordance with MIL-E-15090, Class 2. Coating the cabinet interiors with chemical film in accordance with MIL-C-5541, Class 1A, shall be permitted in lieu of painting.

3.4.29 Component Parts. Component parts in the power supplies may be mounted "piggyback." They may have a cum. total of both leads that exceed 1", but in no event shall it be greater than 1.625". A pliant buffer material shall not be required on circuit boards, when conformal coated with type UR material.

3.5 Antenna Systems. The PC shall be capable of operations with two types of antenna systems. The two types of antennas are the electronically steerable array (ESA) or a mechanically rotating antenna. Selection of the antenna type shall be selectable from the PC front panel.

3.5.1 ESA Antenna. The ESA antenna is a circular array nonrotating assembly. Positioning of the antenna beam shall be accomplished through independent control interface that supplies digital data necessary to control the azimuth position of the ESA antenna. This control data shall be generated in the PC allowing jump or rotation scanning of the ESA antenna. Technical characteristics shall be as specified in DOD-AIMS-64-600.

3.5.2 Rotating Antenna. The rotating IFF antenna can be a linear array mounted upon a pedestal, an integral IFF antenna, or mounted "piggy back" onto a radar antenna. Bearing information (analog synchro signals) is fed from the antenna pedestal to the PC. Technical characteristics shall be specified in MIL-A-28737.

3.5.3 Interface Design. The AN/UPX-24(V) shall interface with many peripheral equipments (radars, switchboards, display consoles, computers, and so forth). The design of the AN/UPX-24(V) shall not require modifications to the peripheral equipment design parameters.

3.6 Interrogator. The design of the central IFF system shall include at least one, but preferably two, AIMS type interrogators. The interrogator equipment is the AN/UPX-27 which is specified in MIL-I-28763.

3.6.1 Variable RF Power Output. One of the requirements for the IFF system is to provide programmed selectivity of RF power output. The purpose is to provide greater and lesser power in the RF challenges on a sector or directional basis to match the variable ranges of tactical operation.

3.7 Crypto Equipment. The equipment to be included in the central IFF system design is the KIR-1A which will be Government-furnished equipment (GFE).

3.8 PC. All functions of the PC shall be provided through the entire range from 0 miles to 300 miles range. A capability for delaying the IFF displayed target from no delay through 45 microseconds shall be provided. If practicable, this type of delay adjustment shall be independently controlled for each of the radar synchronization units. A capability for bypassing the PC main unit and controlling the operation of the interrogator directly from the CI units shall be provided. This capability shall be independent of the PC main unit and operate whether or not the PC main unit is energized. This by-pass feature shall include the interfacing with radar triggers, synchro data, and timing synchronization.

3.8.1 Antenna Control. The control signals for positioning the ESA antenna shall be provided to achieve the interrogation patterns required by the individual operating requests and control of the PC. Continuous scan, sector scan, and searchlight beaming are typical operations. The requirements for operation of the antennas shall be as specified in MIL-A-28737 and DoD AIMS 64-600.

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3.8.1.1 Antenna Steering Signals. Ten-bit sequential steering signals shall be provided by the PC for directing the electronically steered antenna system OE-120/UPX. The speed of rotation of the directive beam shall vary from 0 revolutions per minute (RPM) to 60 RPM. Upon receipt of an externally generated pop-up steering signal (from NTDS or other system), priority shall be given to the request and passed to the antenna directly without continuing the sequential rotation, except when Mode 4 evaluation is in progress. When the externally generated request signals are removed, the internally generated sequential signals shall resume the control of the antenna beam positioning.

3.8.1.2 Synchronization of the Steering Signals. The PC shall be capable of providing the antenna steering signals both synchronously and asynchronously with the repetition frequency of the interrogation cycle. Selection of either method shall be through simply made internal control or link adjustments available to the maintenance personnel. This selection shall be made at the PC only as this choice is not an operator function. During synchronous operation, the steering signals are provided during the dead time of the interrogation cycle. The rotational rate of the sequential steering shall be determined by the repetition frequency of the interrogations. During asynchronous operation, the steering signals are provided at a continuous rate and not regulated by the repetition frequency of the interrogator. The rate control equipment requirement is specified in 3.8.1.1.

3.8.2 Interrogation Control. The PC shall provide the interconnection and control to the interrogator to select the challenge modes, determine the RF power output, and establish the interface patterns. The technical characteristics for the interrogation control shall be as specified in DoD AIMS 65-620 and DoD AIMS 64-402.

3.8.2.1 End of Sweep Trigger. An end of sweep trigger output suitable for interfacing with the AN/UPX-27 interrogator shall be provided. This trigger is nominally a positive pulse of 5 volts (V) to 20 V magnitude into a 75 ohms resistive load. The trigger shall operate the video inhibit circuit when the repetition frequency of the AN/UPX-24(V) increases beyond the receiver gate length of the interrogator. This condition can easily occur in the main operating position of the sector controls. The PC shall function with the resultant countdown that occurs.

3.8.3 Reply Signal Processing. The reply signal processing circuits shall accept the IFF replies from the interrogator receiver and process the replies to produce a digital report and PPI display signal for each reply received during each interrogation cycle. The central processor shall distinguish between valid and nonvalid signals and assign to each reply (received during successive interrogation cycles) the altitude, range, azimuth, and identity codes, and produces a signal position report for storage or display or both. PPI display signals shall be real time and the report message specified in (a) through (h) shall be performed during the signal processing cycle:

a. Signal quantization and standardization
b. Garble detection and indication
c. Defruiting
d. Decoding
e. Reply validation (including special signal evaluation)
f. Reply range indication
g. Reply azimuth indication
h. Alarm generation

NOTE: The sequence (a) through (h) is not intended as the required processing sequence. The contractor should determine the sequence for optimum performance.

3.8.3.1 Signal Quantization and Standardization. Signals shall be detected and identified when the signals are small in magnitude and less than average noise peaks by a ratio of at least one to two. The circuitry shall

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identify that a signal is present in the noise, and deliver the signal to the standardizing circuits. The standardizing circuits shall provide a standard output for all signal levels from the minimum to the maximum (saturated) video supplied by the receiver of the interrogator.

3.8.3.2 Garble Detection and Indication. When a garble occurs, an indication to that effect shall be included in the data to be stored or displayed or both. When garble occurs because of proximity of more than two replying targets, only the first and the last bracket decodes are required for display. The identity of the number of targets producing the garble is not required. Guidance for garble sensing is the Degarbling paragraph of DoD AIMS 65-619.

3.8.3.3 Defruiting. The PC shall remove asynchronous replies from the video signals supplied by the receiver of the interrogator. The method and the place in the sequence of processing operations shall be chosen to give the greatest reduction of the asynchronous replies at the lowest circuit complexity and dollar costs. The defruiting circuits can be combined with validation circuits or other functions of the PC. Defruiting schemes that pass or generate false codes shall identify that a false code may exist. The defruiting cannot remove a whole interrogation cycle from the processing or the processed outputs each time a new mode of interlace is introduced.

3.8.3.3.1 Real Time IFF Data. There shall be defruited output (in real time) of the incoming IFF video. This output shall represent the operations of the IFF system at whatever azimuth the IFF antenna is pointing. This real time output shall be available whether or not the PC main unit is operating. To achieve this capability, a PC by-pass unit may be used.

3.8.3.4.1 Passive Decode Symbology. The design for passive decoding and displaying symbols produced in the AN/UPX-24(XN-2) shall be used in the AN/UPX-24(V) system unless a superior method of symbol generation and shape exists. Approval to change from the method utilized in the AN/UPX-24(XN-2) shall be obtained from the procuring contracting officer.

3.8.3.5 Reply Validation. The PC shall compare reply information received on successive interrogations in corresponding modes, and shall employ a criteria check to establish the validity of the information contained in or associated with the replies before passing this information onward to storage or display. For SIF codes, the criteria check will require that at least two reply codes compare identically and that at least one of the codes shall be nongarbled. For Mode 4, requirements for enemy admission and friend rejection (see 3.4.2, items a and b) shall be complied with.

3.8.3.5.1 Mode 4 Evaluation. The Mode 4 evaluation shall be an automatic decision procedure that considers the operational environment (density of targets and interrogations) and displays the selected decision in a readout indicator as specified in (a) and (b):

a. For Mode 4 evaluations initiated from a CI the results shall be displayed on a readout indicator as:

   (1) FRIENDLY (GREEN or RED): Green for targets conforming to target reply criteria; Red indicating a hostile environment. A Mode 4 symbol shall be displayed on the console for each FRIENDLY target.

   (2) HOSTILE: For targets indicating jamming characteristics.

   (3) UNKNOWN: For targets not replying or not providing sufficient replies to reach FRIENDLY or HOSTILE decision.

b. For Mode 4 evaluations initiated via the NTDS computer interface the results are reported as follows:

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(1) FRIENDLY: For targets conforming to target reply criteria.

(2) UNABLE TO EVALUATE: For targets replying but unable to meet FRIENDLY or HOSTILE criteria.

(3) HOSTILE: For targets meeting density tests criteria indicating jamming characteristics.

(4) NO MODE 4 RESPONSE: For targets not responding to Mode 4.

3.8.3.5.2 PC Overloading. When in Mode 4 evaluation only, the PC shall not overload when the number of targets in a given interrogation is 30 or less.

3.8.3.6 Range Indication of the Reply. The signal processing shall be capable of producing a single value for the range of each target established during each illumination period.

3.8.3.7 Azimuth Indication of the Reply. The signal processing shall be capable of producing a center of azimuth for each target by determining start of target and stop of target and indicating the mid-point between. It is recommended that all mode replies be utilized in determining this center of azimuth.

3.8.3.8 Alarm Generation. The several alarm outputs specified in DoD AIMS 65-619, for emergency and communication failure replies shall be provided in the central processor. Additional alarm circuits for monitoring proper operation of the central IFF system may be included but may not utilize the alarms for the emergency replies.

3.8.3.8.1 Remote Fault and Power Indication. Capabilities shall be provided to indicate at all CIs when a BITE-detected fault occurs in any PC processor or functional unit or when a PC BITE-detected failure occurs in the interrogator receiver or any combination thereof. Automatic shutdown capability shall be provided at the PC in case of an over temperature, over voltage, under voltage, or over current conditions.

3.8.4 Sensing of Multiple Replies. The replies from targets that are separated in slant range by less than 3.15 nautical miles will be received within a 40 µs period. Such group replies can mutually interfere with each other. Such a condition is a potential garble and in addition, an IDENT reply may be received with certain data pulses spaced to meet the bracket pulse spacing requirement. The performance of the processing functions shall consider the following: IDENT replies, overlapped replies, interleaved replies, closely spaced replies.

3.8.4.1 IDENT Replies. If a nongarbled reply contains the SPI pulse and the data pulse C2 (spacing similar to bracket pulse spacing), provision shall be made in the processor logic circuitry so that neither a bracket nor a garbled reply is indicated.

3.8.4.2 Overlapped Replies. When two or more replies are received with any of the data or bracket pulses of a reply overlapping the data or bracket pulses of another reply, the processor shall perform the functions in (a) through (d):

a. Indicate the existence of two replies based upon sensing the bracket pair for each reply provided the leading edges of the first bracket pulses are separated by at least 0.8 microseconds.

b. Provide range and azimuth indication for each reply

c. Provide an indication from degarbling circuitry to designate potentially garbled replies.

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d. Provide the altitude or identity code associated with each reply

3.8.4.3 Interleaved Replies. The signal processor shall decode and process both of the interleaved replies as ungarbled replies provided the interleaved pulses do not fall inside the garble detection gates.

3.8.4.4 Closely Spaced Replies. Closely spaced replies are two nonoverlapped and noninterleaved replies with the first bracket pulse of each reply between 21.1 microseconds and 40.6 microseconds apart. Each reply shall be sensed separately without garble indication.

3.8.5 Reply Processing Rate. The signal processor shall be capable of processing all replies received at the rates specified in (a) through (e):

a. Up to 30 replies per interrogation cycle. Readout of codes may be restricted under this condition

b. Any number less than 30 replies closely spaced by nonoverlapping and noninterleaved

c. Two overlapped or interleaved replies with leading edges of the first bracket pulse of each reply separated by at least 0.8 microseconds

d. Two overlapped or interleaved replies followed by another reply separated from the two by at least 25 microseconds from the leading edge of the last pulse of the two over lapped or interleaved replies

e. When reply rates exceed the conditions in a, b, c, and d, no errors shall be introduced into, or data lost from the replies already in process before the overload condition started, nor when the overload condition is terminated

3.8.6 Total Target Capacity. The system shall provide total target storage capacity of at least 256 targets, with a goal of 400 targets through simple expansion features.

3.8.7 Range Resolution and Range Error. The range resolution of the system shall be at least 200 yards and the range error produced by the PC circuits shall never exceed 200 yards.

3.8.8 Azimuth Indication and Azimuth Error. Azimuth indication of 1024 incremental position identifications shall be provided (approximately 0.35°) and the azimuth error shall not exceed one increment, or 0.35°.

3.8.9 Processing of Special Replies. Provisions shall be made in the signal processing to recognize and identify military and civilian emergency replies, communication failure replies, X pulse, and SPI and I/P replies. Special tags shall be added to the processed reply message, as required, during the processing of these replies.

3.8.10 Data Processing Rate. The output signals for the message report for each reply for storage or display, or both, shall be available for a given target within two interrogation cycles following the stop of target indication.

3.8.11 Target Report Message. The target report message shall be determined largely by the memory capability of the storage unit; however, requirements of the existing AIMS target reports of DoD AIMS 65-620 and the requirements for the NTDS interfacing MIL-STD-1397, Type B should be considered in determining the target report message construction.

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3.8.11.1 Target Reports. The output of the PC to the computer type interface of the shipboard NTDS and weapon control systems shall contain a complete target report plus communication status and interrupt signals necessary for the control of the AN/UPX-24(V) through the shipboard computers. By definition, the target report contains all the IFF data, target azimuth and range, Mode 4 evaluation, and pop-up target data.

3.8.11.2 Pop-Up Target Operation. When the PC pop-up target is requested via the NTDS computer interface, the PC shall interrupt operations to perform instant challenge, decode, and readout functions on a selected target. The priority of requests shall be that the NTDS computer interface requests shall be honored above CI requests except when a CI Mode 4 evaluation is in process, the NTDS computer request shall not interfere with the CI's overlapping request in process but will supersede all other CI requests awaiting processing. The NTDS computer initiated pop-up target information shall be delivered to the NTDS computer interface and shall not be stored in the PC memory.

3.8.11.3 CI Popup Function. The CIs may request target information on selected targets for readout from stored information in the PC. Targets are selected using the console controls and the information is read out on the CI display windows.

3.8.12 Target Trails. Position data of all target reports may be retained in memory to provide trails of displayed targets. Up to three small position dots may be displayed behind the current symbol indicating current position.

3.8.13 Additional and Optional Features. The features of signal processing specified in 3.8 represent minimum performance requirements. Several features specified in DoD AIMS 65-620 can be included. There are similar features in other automatic systems already designed for the AIMS program. The costs of circuitry complexity and dollar amounts should be considered and the best trade-offs solicited.

3.9 Control Box and Active Readout. The control functions that shall be supplied at the multiple operating positions shall be similar to at least those specified in the controls, indicators, and panel layouts paragraph of DoD AIMS 65-619. Additional controls may be necessary (or desired) to achieve full benefits of the system design. The active readout (numeric display unit) should provide at least the features specified in the SIF/Altitude readout units paragraph of DoD AIMS 65-619.

3.10 Buffer Storage Unit. The buffer storage unit shall be capable of storing total target data for at least 256 targets, and of supplying (upon suitable electrical commands) the required outputs for the display generator unit or the NTDS interface unit, or both. Range resolution from the buffer storage shall be constant with the system capability of one-tenth mile resolution. Azimuth, or bearing information shall be in 1024 increments (10 bitword) for 360° coverage. Mode information, code information, passive decodes, bracket decodes, garble tags, and so forth shall be included through economical use of storage elements. Readout shall not destroy the stored information. The stored information shall be updated as often as new information is produced by the interrogator and PC. The reports for pop-up or for Mode 4 evaluation targets shall have priority in the delivery to the computer interface.

3.11 Display Generator. The display generator requests outputs from the buffer storage unit and converts the data into signals suitable for the readout displays. The range and azimuth gating method of selecting the targets to be so displayed shall be similar to the active decoding features outlined in DoD AIMS 65-619, or preferably some easier operating method such as a light pencil or other target selecting procedure.

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3.12 NTDS Interface Equipment. This unit shall provide the translation of the IFF information into language suitable for the NTDS system and also translate the NTDS control words into suitable IFF control signals. Technical guidance for providing this capability is specified in MIL-STD-1397.

3.12.1 Computer Interface Capability. Interface capability shall be designed to accept the type of interface signal specified in (a):

a. NTDS fast interface: -3V; 167,000 words per second, maximum

3.13 Distribution System. Distribution of the IFF information to the multiple operating positions can be either through existing switchboards installed aboard ships, or can be directly wired and by-pass the switchboards. The choice should be made on the basis of the data to be distributed. Replacement of switchboards to provide greater input and output capacity shall not be required. Installation of a great many cables just to overcome the problems of switchboard capacity shall not be acceptable. A simple distribution system shall be designed into the central IFF system that will make use of the switchboards for real time information and direct wiring of automatic data to computers of the NTDS, CATCC, and similar systems.

3.14 Display and Display Method. The existing display consoles will be varied. Some will be PPI repeaters; some will be NTDS consoles. The display signal shall be independently selectable at each operating position. Synthetic displays shall be target position, alphanumeric code, altitude data, and emergency signals. The alphanumeric code and altitude data may be displayable only on the CI. Standard IFF video signals shall be bracket decoding, and passive decoding similar to bracket decoding with signal enhancement (such as pulse stretching). These standard IFF signals shall appear in real time in the characteristic IFF arc displayed slightly delayed in range from the radar target display.

3.14.1 Display Outputs. When switchboard distribution is utilized, at least three and preferably four parallel outputs will be required for isolated feeding of the separate switchboards.

3.15 Workmanship. Workmanship for the system shall conform to Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government Verification. All quality assurance operations performed by the contractor will be subject to Government verification. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

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4.1.1.1 Failures (see 6.4). The contractor shall, at no additional cost to the Government, report to the Government all failures and deficiencies which occur during Acceptance Testing of the AN/UPX-24(V) systems (including the workmanship screen tests required in para. 4.2.2) and Acceptance Testing of all spares. Pattern failures (as defined by MIL-STD-781) which occur during these tests shall be identified and reported to the Procuring Activity. In the case of pattern failures, the contractor shall analyze the failure or discrepancy, determine the corrective action required, and implement and demonstrate the effectiveness of the corrective action to the satisfaction of the Government.

4.1.2 Quality Assurance Terms and Definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Production inspection (see 4.2.1)
b. Inspection of preparation for delivery (see 4.2.3)

4.2.1 Production Inspection. Production inspection shall be performed in accordance with the tests listed in MIL-E-16400 except as specified in 4.2.2 through 4.2.8. Testing shall include those tests listed in Table I.

Table I. Production Inspection.

Examination and tests	Test Requirement	Method
Marking	3.4.15	4.2.1.1
Parts and materials	3.4.6	4.2.1.1
Workmanship	3.15	4.2.2
Performance	3.1.1	4.2.4
Painting	3.4.28	4.2.1.1

4.2.1.1 Production Inspection Test Methods. Production inspection shall be made on every item offered for delivery. Production inspection shall comprise such examinations and tests which will prove the workmanship and reveal omissions and errors of the production process such as, functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls or adjustments, and tests which detect hidden defects of materials. Unless otherwise specified herein, the production inspection tests shall be performed in accordance with MIL-E-16400.

4.2.2 Workmanship Screen. Vibration and temperature cycling shall be performed on each IFF equipment. Vibration shall be performed prior to temperature cycling. The screen shall be performed prior to the acceptance testing. The vibration may be performed at the module or drawer level. All the hardware including cables and connectors shall be exposed to vibration. Replacement of failed modules (PCB, power supply, etc.) during the Acceptance Test will require repeating two full cycles of the temperature curve as shown in FIGURE 5, or replacing a failed module with one that has passed module/SRA screening.

a. Module SRA/Screening. Each SRA/module shall undergo 10 rapid thermal cycles with a minimum rate of temperature change of 15°C/minute, non-operational, between -54°C and +71°C.

b. WRA Manufacturing Screening. Manufacturing screening shall be performed employing thermal cycling with a minimum rate of temperature change of 10°C/minute (in accordance with Figure 5) and random vibration (6 grms, 20-2,000 Hertz, in accordance with Figure 4) in one axis for a total accumulated time of 10 minutes in the axis most susceptible to vibration. The

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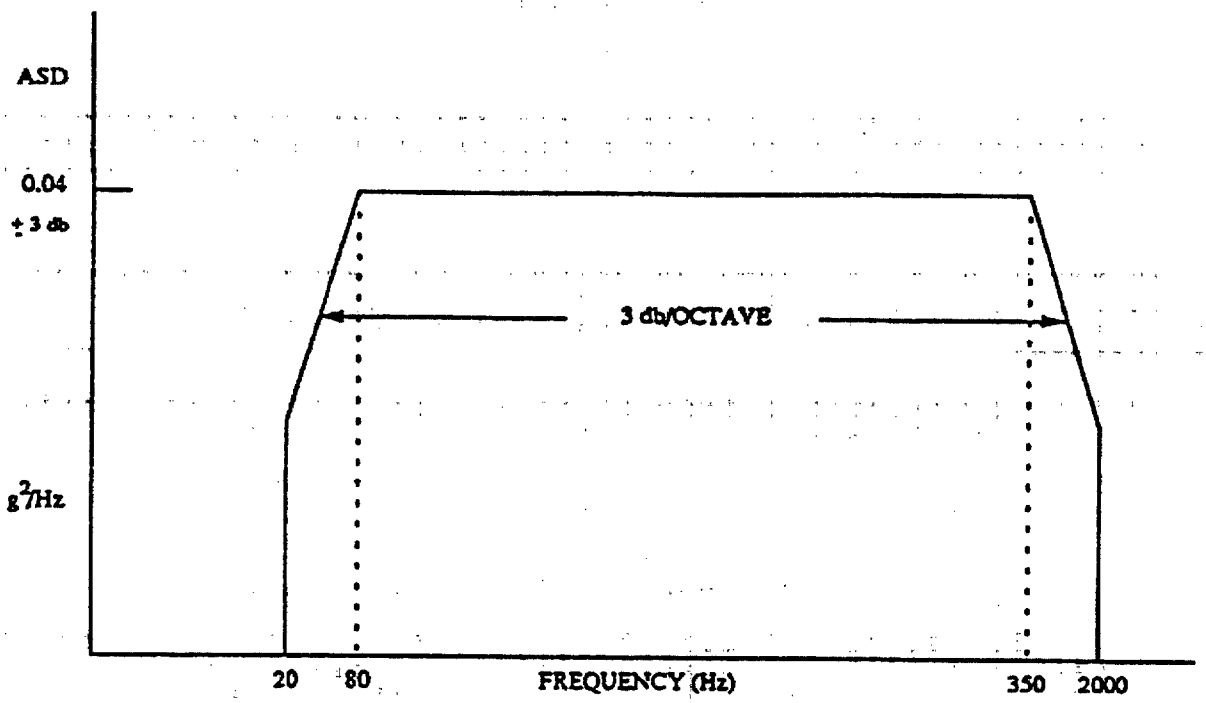


FIGURE 4. Random vibration curve

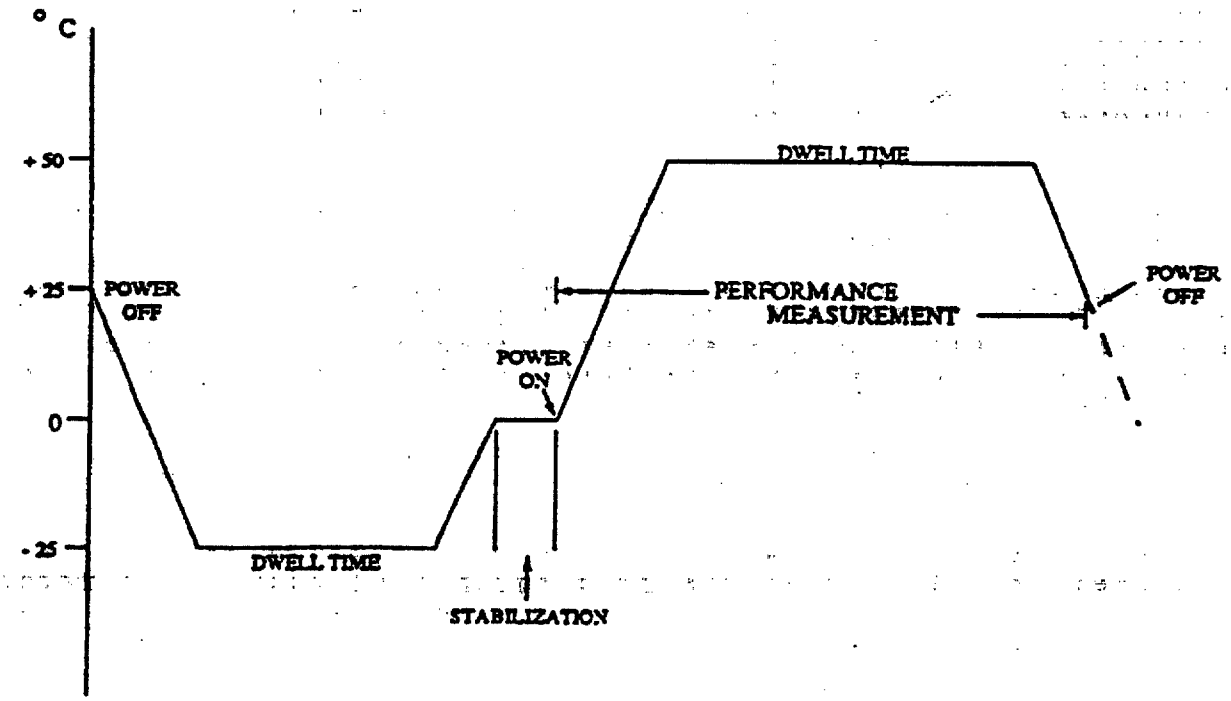


FIGURE 5. One cycle of temperature curve (not to scale)

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temperature cycling for each WRA shall be 40 failure-free operating hours in an 120 operating hour window. The random vibration period shall be performed at ambient before any thermal cycling.

4.2.3 Inspection of Preparation for Delivery. Inspection shall be conducted to ensure conformance with the requirements of Section 5.

4.2.4 Performance Testing. Performance testing shall be performed by the contractor to verify performance requirements outlined in Section 3. The contractor shall use the most current approved production acceptance test procedures.

4.3 Quality Control. In addition to MIL-Q-9858A requirements, the contractor shall provide and maintain a corrective action and disposition system for nonconforming material in accordance with MIL-STD-1520, and a supplier quality assurance program in accordance with MIL-STD-1535, for all supplies and services provided under this contract. The contractor shall maintain a program acceptable to the Government, for the investigation, corrective action and timely response to Quality Deficiency Reports issued by the using activity. In addition to requirements specified above, the following documents shall be applicable to electrical/electronic equipment to be delivered under this contract:

a. MIL-STD-454, Standard General Requirements for Electronic Equipment, less the following:

(1) Requirement No. 5

(2) Requirement to conduct First Article Testing of PC Boards per MIL-P-28809 Paragraph 3.2 in requirement No. 17.

(3) Requirement No. 64

b. MIL-STD-275, Printed Wiring for Electronic Equipment.

c. MIL-P-55110, Printed Wiring Boards.

d. MIL-STD-2000, Standard Requirements for Soldered Electrical and Electronic Assemblies, 16 January 1989, Task F with the following modifications:

(1) The following items shall be exempt from the flowdown provisions of MIL-STD-2000, Paragraph 5.4.1:

ITEM DESCRIPTION	PART NUMBER
DISPLAY, 7 SEGMENTS	865341
BEZEL, 3 DISPLAY	169131
BEZEL, 2 DISPLAY	169132
BOARD, 3 DISPLAY	169143
BOARD, 2 DISPLAY	169144
PANEL, CONTROL INDICATOR	169121
SWITCH	865361-001 THRU -107
READOUT, INCANDESCENT	865362
PANEL, CTRL INDICATOR	169371
PANEL	169130

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(2) The contractor may pre-tin component leads in accordance with MIL-P-28809A Paragraph 3.4.3, "Component Leads and Wires", prior to use in assembly (soldering) in lieu of employing the steam-aging solderability testing required in accordance with MIL-STD-2000, Paragraph 4.4, "Solderability."

e. MIL-STD-1686, Electronic Discharge Control Program.

Any provision of the above documents (a) through (e) relating to design, materials of construction as parts selection shall not apply to previously designed equipment and therefore redesign shall not be required as a result of the inclusion of the above documents. Manufacturing/processing tools and materials such as solder, flux and solvents are not included in this exception.

5. PREPARATION FOR DELIVERY

5.1 Preservation, Packaging, Packing, and Marking. The equipment, accessories, technical publications, and repair parts shall be preserved, packaged, packed, and marked as specified in the contract or order.

6. NOTES

6.1 Intended Use. The central IFF system is intended to transmit AIMS IFF interrogations, to receive and process the replies and to distribute the processed information to various operating and display positions without reliance on radar synchronization or require the use of other shipboard equipments.

6.2 Ordering Data. Procurement documents should specify the following: Title, number, and date of this specification

6.3 PPI Displays and Radar Distribution Systems. For economic reasons, the use of existing PPI displays and radar distribution systems shall be considered.

6.4 Failure. The failure definitions of MIL-STD-781 shall apply throughout this specification.

6.5 Subject Term (Keyword) Listing.

ATCRBS, Air Traffic Control Radar Beacon System
BITE, Built in Test Equipment
IFF, Identification Friend or Foe
Indicator, Remote Control
Interrogator Set
Monitor, Remote Control
Processor, Unit Central
SRA, Ship Replaceable Assembly
WRA, Weapon Replaceable Assembly

Preparing Activity:
Navy - AS
(Project 5895-N591)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I-29589 (AS)	91/07/26

3. DOCUMENT TITLE

INTERROGATOR SET, AN/UPX - 24 (V)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
COMMANDING OFFICER NAVAL AIR ENGINEERING CENTER SYSTEMS ENGINEERING & STANDARDIZATION DEPT.	(908) 323-2326	624-2326
c. ADDRESS (Include Zip Code) CODE 53 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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